

Control Instruction :

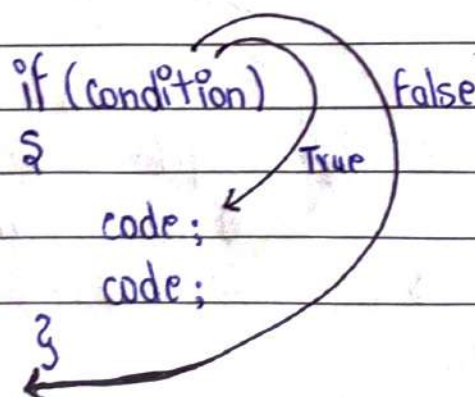
We study control instruction to control the flow of program sequence.

1. Decision control instruction
2. Iterative control instruction
3. Switch case control instruction
4. goto control instruction

Decision control instruction & selection control statement ?

1. if
2. if else
3. ? : (conditional operator)

if



- Every non-zero value is True

- Zero is false

if else

if (condition)

{

code;

code;

}

else

{

code;

code;

}

True

false

- If there is only one code or statement inside if & else block so it is not compulsory to use curly bracket.

Conditional operator

(Ternary operator)

exp 1 ? exp 2 : exp 3 ;

Always condition

True

false

Q → WAP to check whether a given no. is positive or non-positive

Preprocessor directive

#include <stdio.h> → Standard input output header file

int main ()

{

int x;

printf("enter a number=");

scanf("%d", &x);

x > 0 ? printf("positive") : printf("non-positive");

}

↓ or

printf(x > 0 ? "positive" : "non-positive");

logical operator :-

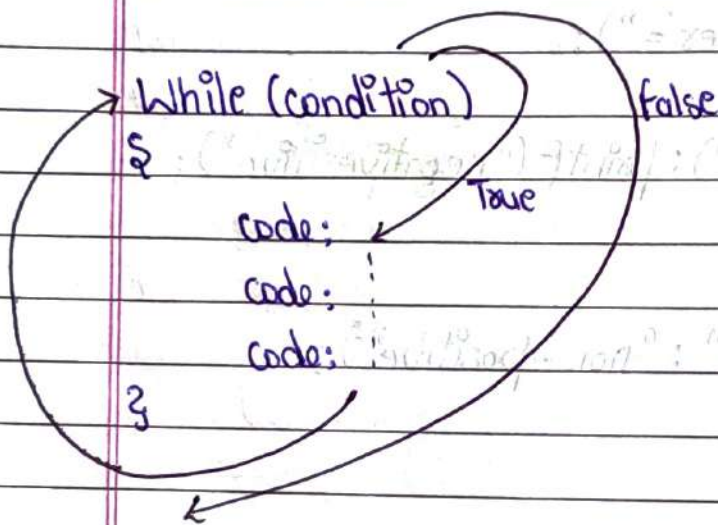
It is mainly used in boolean operation & True or false condition expression

Binary operator :-

It is mainly used for differentiating b/w two variables containing values

Iterative control instruction loop

1. While {entry control loop}
2. do while {exit control loop}
3. for



① → #include <stdio.h>

int main ()

{

int i = 10;

while (i)

{

printf ("%d", i);

i--;

}

}

Every non-zero → True

zero → false

No increment in i

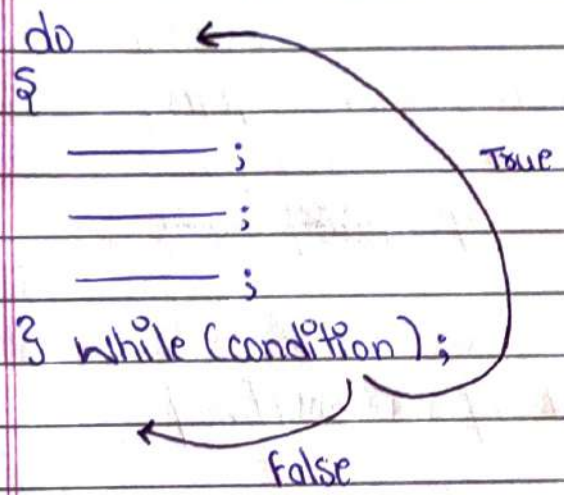
(a) 10 9 8 7 6 5 4 3 2 1

(b) 9 8 7 6 5 4 3 2 1

(c) Infinite

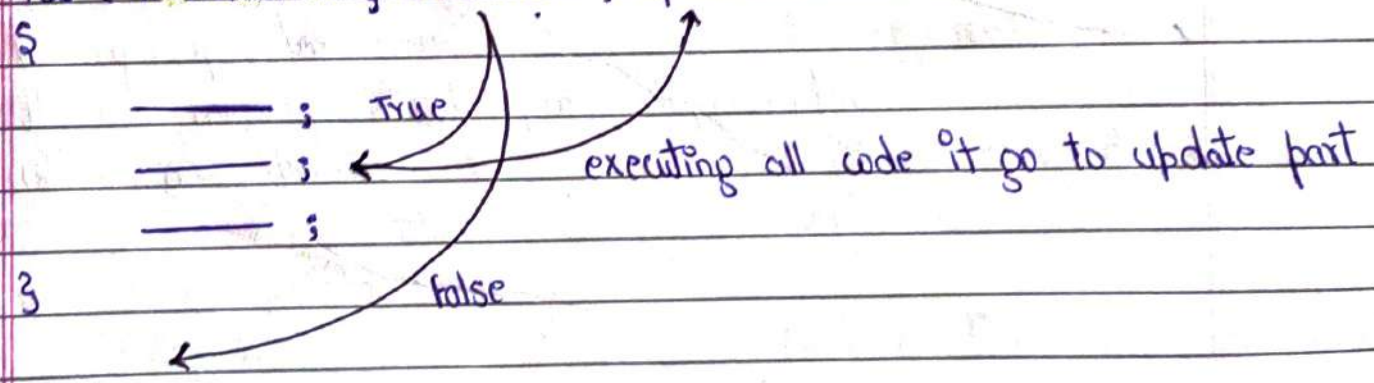
(d) 10 9 8 7 6 5 4 3 2 1 0

do while



for executed once

for (initialization ; condition ; update)



break

break is a keyword

In loop, break is used to terminate execution of loop.
It is also can use in switch case

```
int i = 1;
```

```
while (age <= 100)
```

```
{
```

```
    _____;
```

```
    _____;
```

```
    _____;
```

```
    break;
```

```
    _____;
```

```
    _____;
```

```
    _____;
```

```
    age++;
```

```
}
```

```
←
```

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

(if (age <= 100) { while (age <= 100) { ... } })

Nested loop

While ()

```
{
    for ( ; ; )
    {
        _____;
        _____;
    }
}
```

for (; ;)

```
{
    for ( ; ; )
    {
        _____;
        _____;
    }
}
```

Switch (expression)

{

case constant:

case constant:

default:

}

→ only constant no variable

All constant have to be different

Switch and default are keyword

And constants in case label can only be Integer and character type constant no real constant X